

Performance of a Compritol® 888 ATO Lipid Matrix Tablet to Deliver Sustained Release of Diclofenac Sodium



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INTRODUCTION

Diclofenac sodium is a non-steroidal anti-inflammatory agent (NSAID) with antipyretic and analgesic actions. Diclofenac is used to treat pain, dysmenorrhea, ocular inflammation, osteoarthritis, rheumatoid arthritis, ankylosing spondylitis, and actinic keratosis. It is a highly water soluble salt (50 mg/mL).

Compritol® 888 ATO is a glyceryl behenate, with an HLB of 1 and a melting point 70°C. Available as a fine white powder with a mean particle diameter of 50µm, it is used in tablet processing as a lubricant or as a lipid matrix former for sustained release. The mechanism of drug release is primarily diffusion through the insoluble, non-swellable matrix [1].

Compritol® 888 ATO lipid matrix tablets can be made using straightforward methods. Depending on the properties of the active pharmaceutical ingredient (API), direct compression (DC) / wet granulation (WG) can produce stable sustained drug release.

For highly water soluble APIs other simple processing methods are of interest, including solid dispersion [3], melt extrusion and melt granulation. These processes require an additional step where the drug is dispersed in to molten Compritol® 888 ATO and once cooled the solid dispersion is ground to obtain granules for compression.

The objective of this study is to evaluate the effect of tablet processing methods on Compritol® 888 ATO lipid matrix tablets containing diclofenac sodium.

EXPERIMENTAL METHODS

Compritol® 888 ATO is a Glyceryl behenate NF, synthesized by esterification of behenic acid (C22) and glycerol and therefore consists of a mixture of mono, di and tribehenate of glycerol, the diester fraction being predominant.

All tablets contained 100mg of Diclofenac sodium (provided by Sochibo). The amount of Compritol® 888 ATO varied subject to formulation and processing methods. Diluents included lactose and anhydrous dicalcium phosphate with magnesium stearate providing lubrication.

For direct compression, excipients and API were firstly blended together, followed by addition of the lubricant to the external phase in a second blending step. Sintering treatment on the tablets was carried out at 80°C for 30 minutes in dry heat conditions.

Solid dispersions were prepared by dispersing diclofenac sodium in to molten Compritol® 888. Once cooled the solid mixture was then ground on an oscillant screen. The powder obtained was calibrated on a 630µm sieve; additional tablet ingredients were blended prior to direct compression. All tablets were prepared on an alternative tableting machine (Korsch Ek03).

The contact angle of tablet surface was measured using a goniometer ILMS from GBX instruments. Drug release profiles were assessed by dissolution testing equipped with UV measurement. All stability tests were performed under ICH conditions.

RESULTS AND DISCUSSION

Direct compression of diclofenac sodium with Compritol® 888 ATO produced tablets with good physical characteristics. However, sustained release could not be achieved even with the use of 50% Compritol® 888 ATO by weight in the formulation. The density of the Compritol® 888 matrix is not sufficient to maintain the drug entrapped when exposed to aqueous solution.

Graph 1 shows the drug release from a tablet made by direct compression with 38.5% of Compritol® 888 ATO, 16.5% of diclofenac sodium (diluents being the remaining ingredients). Diclofenac sodium was completely released after 30 minutes.

The application of a sintering process on tablets made by DC (herewith referred to as DC-S) appears to affect the structure of the lipid matrix resulting in a denser formation. Porosity data before and after sintering confirm this postulation. The repartition of Compritol® 888 ATO is also slightly different and a lipid layer is observed at the surface of the tablet.

To study the effect of heat treatment on lipid matrix tablets, tablets were produced using Compritol® 888 ATO powder dyed blue (figures 1 and 2).



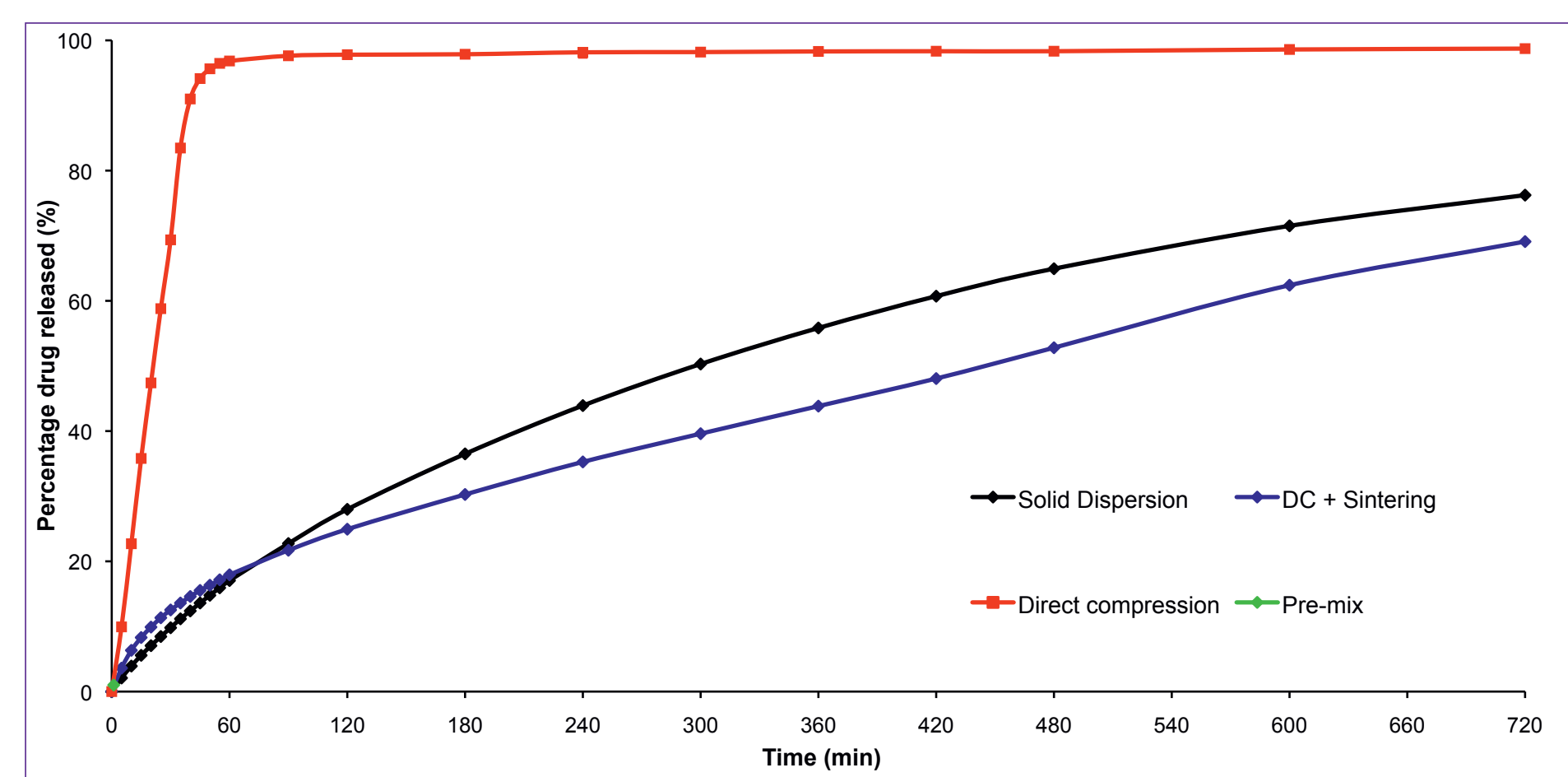
Figure 1 shows a tablet produced by DC (left) and a tablet made by DC-S (right).



Figure 2 shows the cross sectional view of tablets made by DC (right) and DC-S (left).

The process of DC appears to yield an even repartition of Compritol® 888 ATO throughout the matrix and blue particles are visible across the entire tablet surface. A distinct difference in aspect can be seen in the sintered tablet (DC-S); the blue coloration is much darker indicating that the Compritol(R) 888 ATO has melted into the tablet interstices and other ingredients. The cross-sectional view of the tablets (figure 2) shows that Compritol® 888 ATO is distributed homogenously throughout both the DC and DC-S tablets, with the effect of sintering and the melting of the Compritol® 888 ATO producing a much darker coloration throughout the tablet.

Dissolution results for DC-S tablets illustrate the differences in the retardant properties of lipid matrix tablets produced by DC, DC-S and solid dispersion (SD). **These results suggest that sintering may decrease the porosity of the matrix, increasing its retardant capacity resulting in a sustained drug release over 12 hours (see graph 1).**



Graph 1. Percentage drug release from Compritol® 888 ATO matrix tablets made by DC, DC-S and SD.

Solid dispersion is also an effective method to produce a lipid matrix for sustained drug release. Diclofenac sodium was dispersed into molten Compritol® with a ratio 30/70 w/w. After cooling the solid dispersion is ground into a powder, blended with diluents, lubricant and compressed. The tablets contained 38.5% of Compritol® 888, 16.5% of diclofenac sodium and the remaining ingredients were diluents. Again tablets were made with blue Compritol® 888 ATO to evaluate the characteristics of the matrix.

The Compritol in SD tablets is a much darker blue indicative of the melting process. The blue zones appear larger and well-distributed throughout the tablet (figure 3).

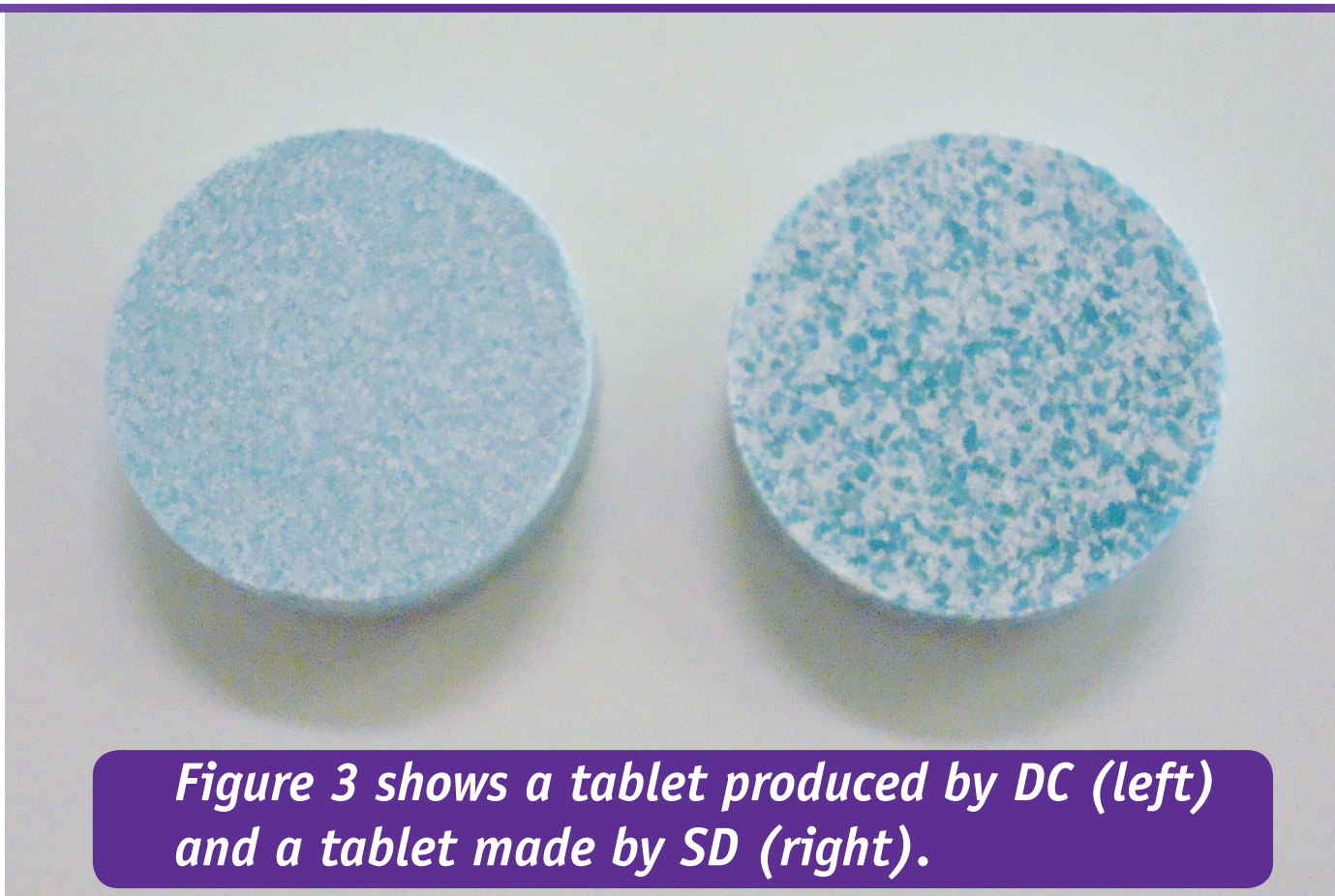


Figure 3 shows a tablet produced by DC (left) and a tablet made by SD (right).

The drug release profile from SD tablets (Graph 1) indicates that the process of dispersing API into molten excipient enables the production of a matrix with good release retardant properties.

To further evaluate the effect of processing methods on tablet characteristics the contact angle of the tablets' surface was measured. The results signify the wettability of the tablet surface (figure 4).

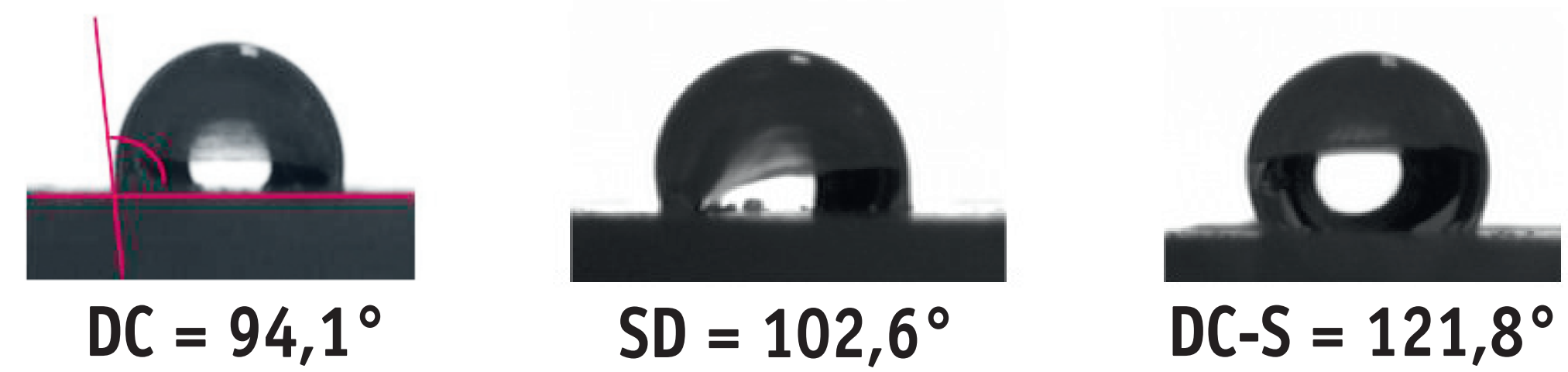
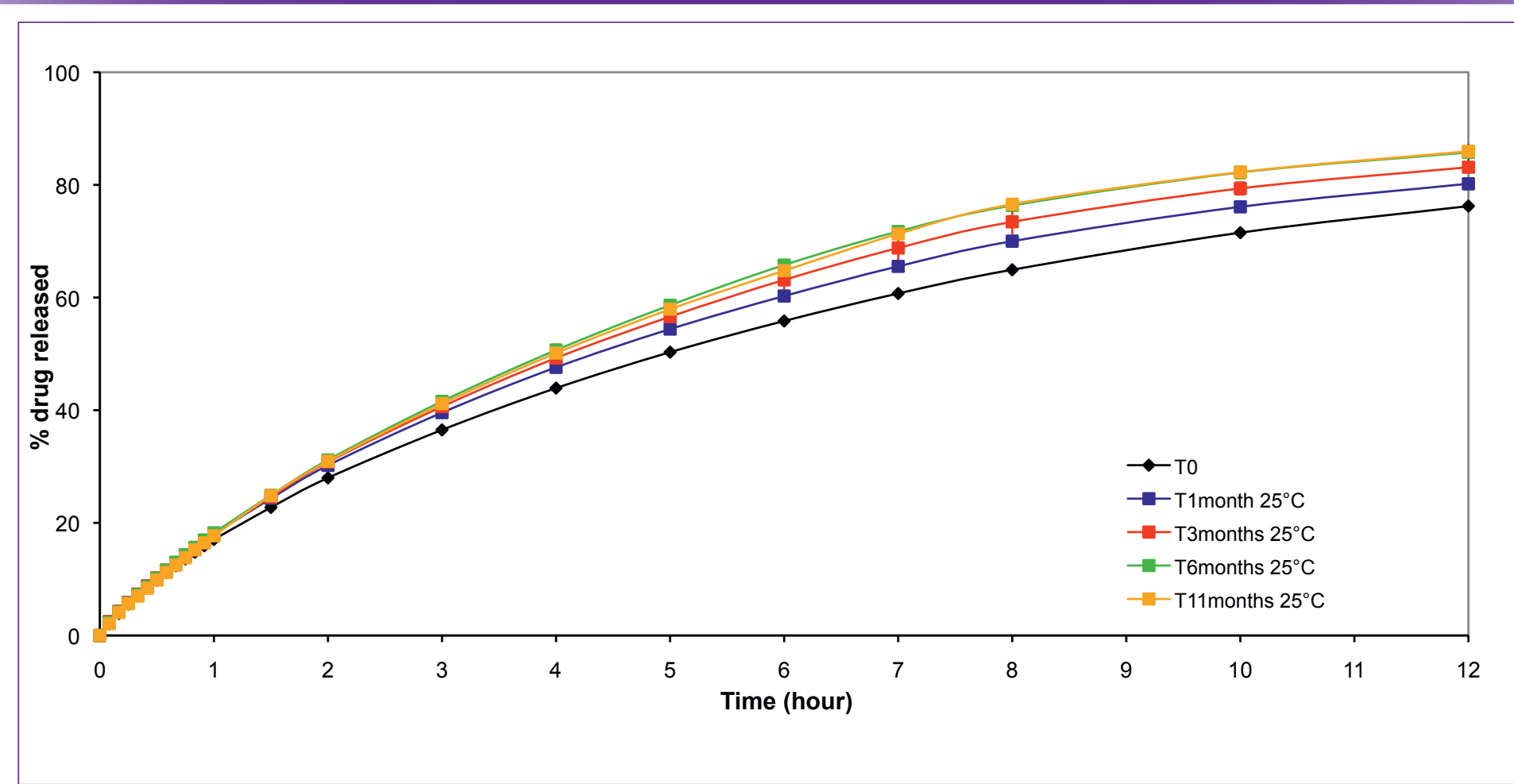


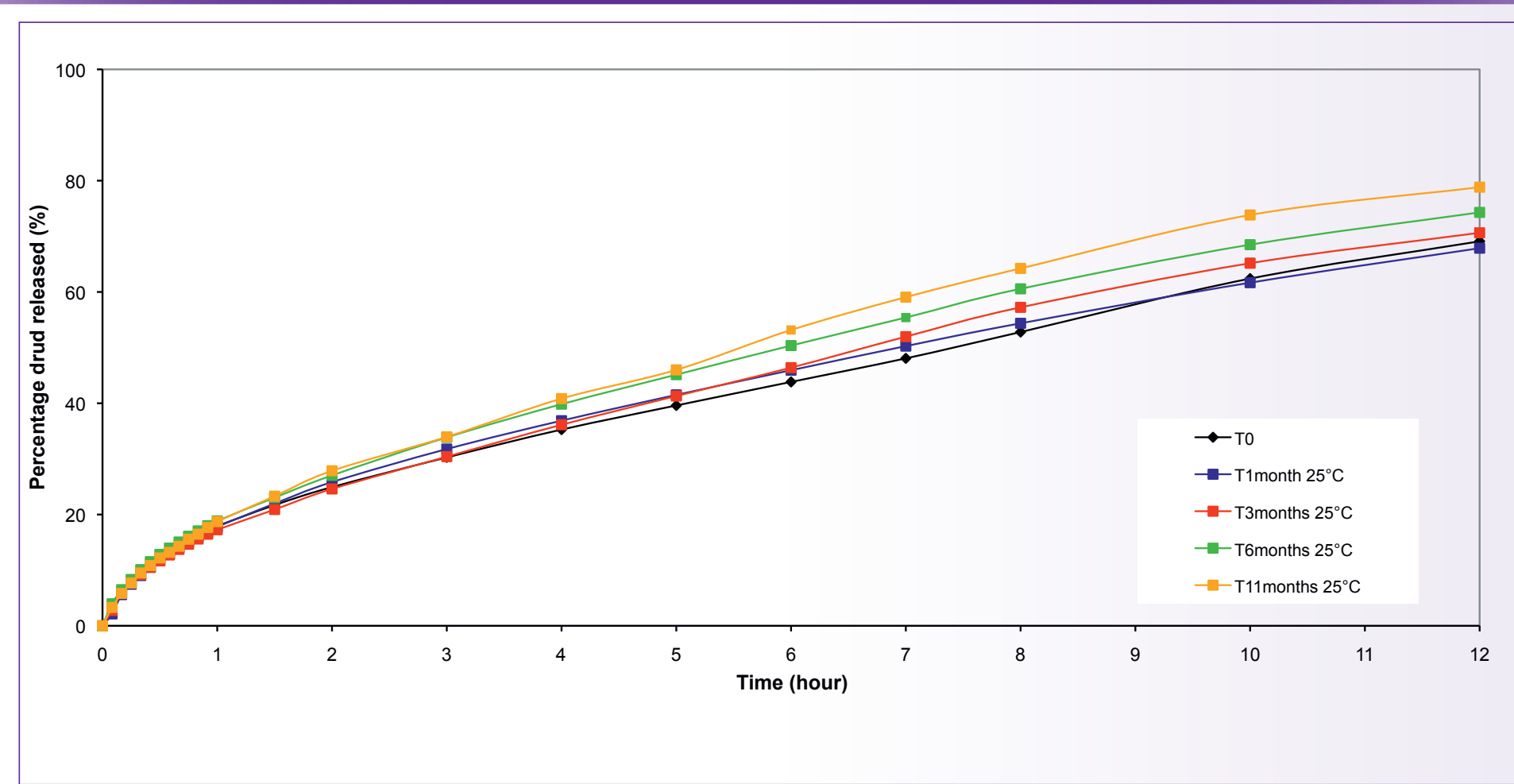
Figure 4 Contact angle for tablets made by DC, DC-S and SD.

Tablets made by DC have the lowest contact angle, 94.1° indicating a greater wettability of the tablet surface compared to DC-S and SD tablets. Solid dispersion increases the surface contact angle to 102.6°, reducing the wettability and increasing the retardant properties of the tablet structure. The sintering treatment produces the highest surface contact angle, 121.8° suggesting the lowest wettability, which confers with the dissolution results.

Graphs 2 and 3 show the results of stability studies for SD tablets and DC-S tablets respectively, indicating stable drug release profiles at (25°C) during 12 months.



Graph 2. Diclofenac sodium-Compritol® 888 ATO tablets produced by SD.



Graph 3. Diclofenac sodium-Compritol® 888 ATO tablets produced by DC-S.

CONCLUSION

Compritol® 888 ATO is a functional excipient that has proven utility for the production of hydrophobic, non-swellable matrix tablets for sustained drug release.

As with all functional excipients, the nature of the API, its water solubility and the dose all influence the choice of formulation approaches with Compritol® 888 ATO. Direct compression can be sufficient for low dose tablets and APIs with medium water solubility. For highly water soluble APIs, like diclofenac sodium, it is important to decrease the porosity of the tablet, generating a denser lipid matrix to sustain the release in aqueous solution. This can be achieved by using straightforward techniques such as solid dispersion or by applying a post-tableting heat treatment such as sintering.

REFERENCES

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